

Smart_Food_Packaging & Cold_Chain_Risk_Analytics

High Risk Packages

2987

Compliance Rate

11

Total Packages

15,000

Cold Chain Failures

30%

Avg Risk Score

60%

Spoilage Rate

72%

Food Category

- ☐ Bakery
- ☐ Beverage
- ☐ Canned Food
- ☐ Dairy

Environmental Monitoring

Packaging & Quality...

Operational Analysis

Food Category by Avg Risk Score

Seafood	0.61
Frozen Food	0.61
Bakery	0.60
Beverage	0.60
Fresh Produce	0.60
Dairy	0.59

Storage Zone by Avg Risk Score



Storage Zone

- ☐ Cold Chain Storage
- ☐ Distribution Center
- ☐ Retail Shelf
- ☐ Transport Vehicle

Food Safety Risk Level Distribution

● Critical Risk ● Medium Risk ● High Risk ● Low Risk

17.95%

19.91%

27.33%

34.81%

Packaging Material by Avg Risk Score

Paperboard	0.61
PET Plastic	0.60
Glass Container	0.59
Vacuum Pouch	0.59
Composite Film	0.59
Aluminum Laminate	0.59
Biodegradable Film	0.59

Packaging Material

- ☐ Aluminum Laminate
- ☐ Biodegradable Film
- ☐ Composite Film

Food Safety Risk Level

- ☐ Critical Risk
- ☐ High Risk
- ☐ Low Risk
- ☐ Medium Risk

Smart_Food_Packaging & Cold_Chain_Risk_Analytics

Avg Shelf Life Left

9

Avg Freshness

57

Avg Temperature

7

Avg Humidity (%)

62

Temperature
Status

- ☐ High
- ☐ Low
- ☐ Medium
- ☐ Very High

Humidity
Status

- ☐ High
- ☐ Low
- ☐ Normal

Storage_Zone

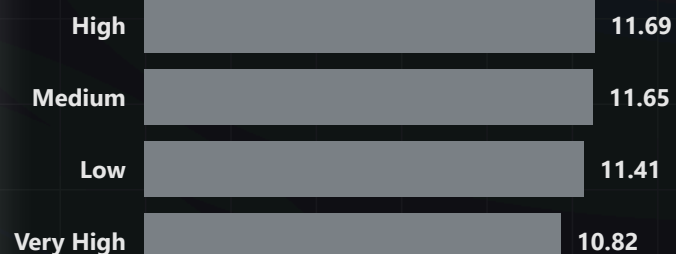
- ☒ Cold Chain S...
- ☐ Distribution ...
- ☐ Retail Shelf
- ☐ Transport Ve...
- ☐ Warehouse

Executive Overview

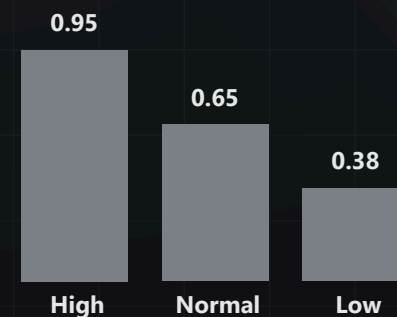
Packaging & Quality...

Operational Analysis

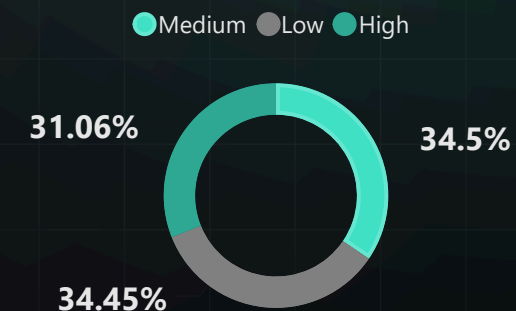
Temperature vs Shelf Life



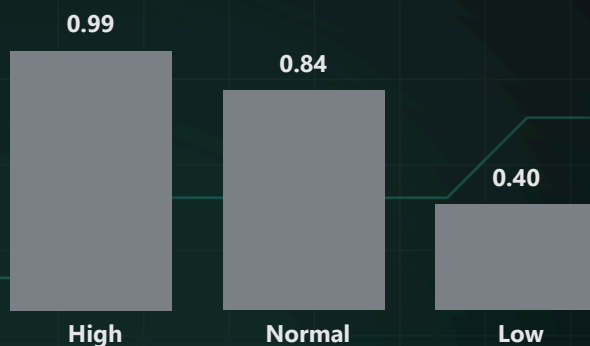
Humidity vs Risk Score



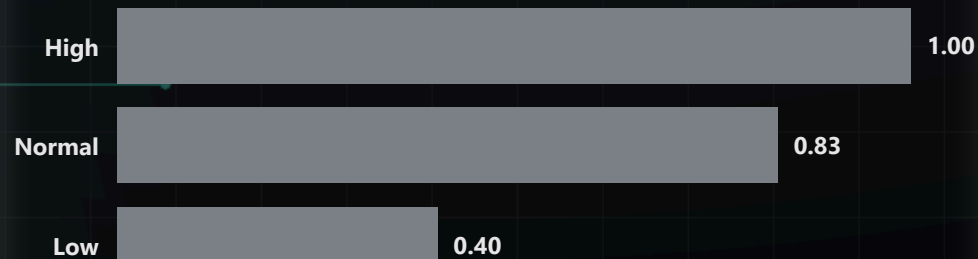
Light Exposure vs Freshness



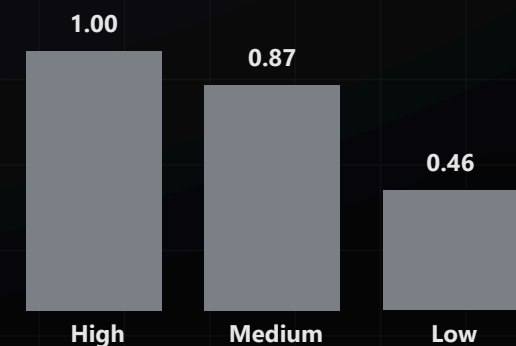
Oxygen Concentration vs Risk



Carbon Dioxide vs Risk



Volatile Gas vs Risk



Smart_Food_Packaging & Cold_Chain_Risk_Analytics

Avg Sensor Health

87

Damage Rate

55%

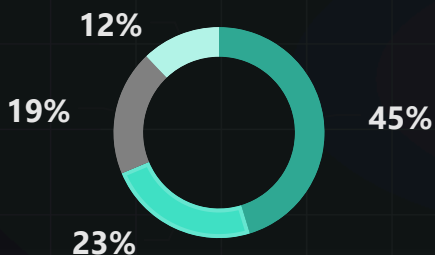
Executive Overview

Environmental Monitoring

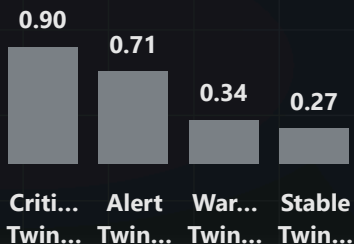
Operational Analysis

Surface Integrity Distribution

Flawless ... Seal Defect Surface D... Contami...



Digital Twin State by Risk Score



Packaging_Material

- Aluminum Laminate
- Biodegradable Film
- Composite Film
- Glass Container
- Paperboard
- PET Plastic
- Vacuum Pouch

Surface_Integrity_Status

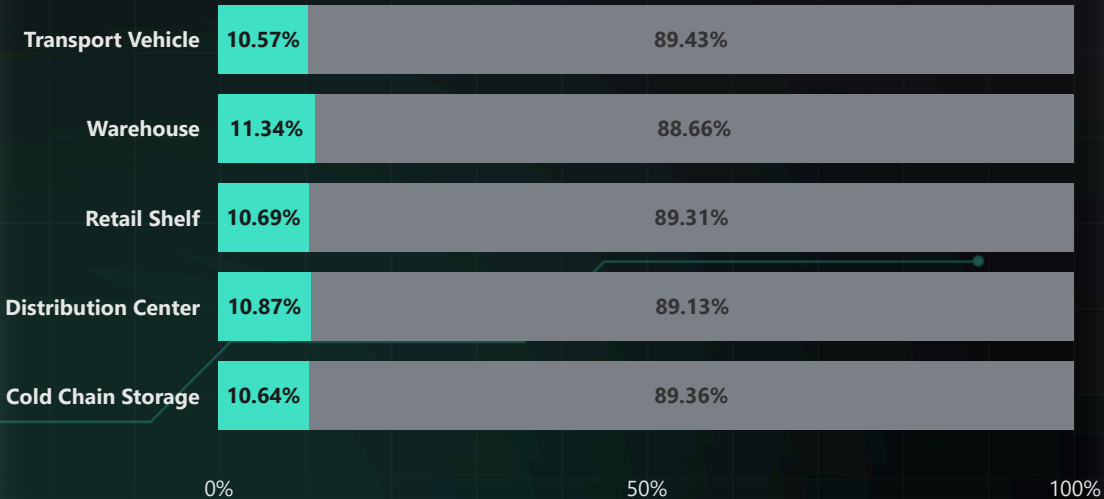
- Contamination Defect
- Flawless Package
- Seal Defect

Digital_Twin_State

- Alert Twin State
- Critical Twin State

Compliance Status by Storage Zone

Compliance_Status Safe Unsafe



Packaging Material vs Food Safety Risk



High Risk Packages by Storage_Zone



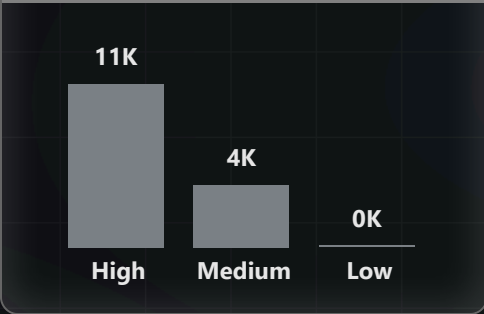
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Executive Overview

Environmental Monitoring

Packaging & Quality...

Transport Condition vs Spoilage Probability



Storage_Zone	Critical Risk	High Risk	Low Risk	Medium Risk	Total
Cold Chain Storage	1029	618	516	797	2960
Distribution Center	1006	618	535	823	2982
Retail Shelf	1042	601	522	829	2994
Transport Vehicle	1089	586	530	851	3056
Warehouse	1055	564	589	800	3008
Total	5221	2987	2692	4100	15000

Transport_Condition

☐ Frequent Vibration

☐ High Moisture Exposure

☐ Mild Shock

☐ Stable

☐ Temperature Fluctuation

Package_Size

☐ Bulk

☐ Large

☐ Medium

☐ Small

Storage_Zone

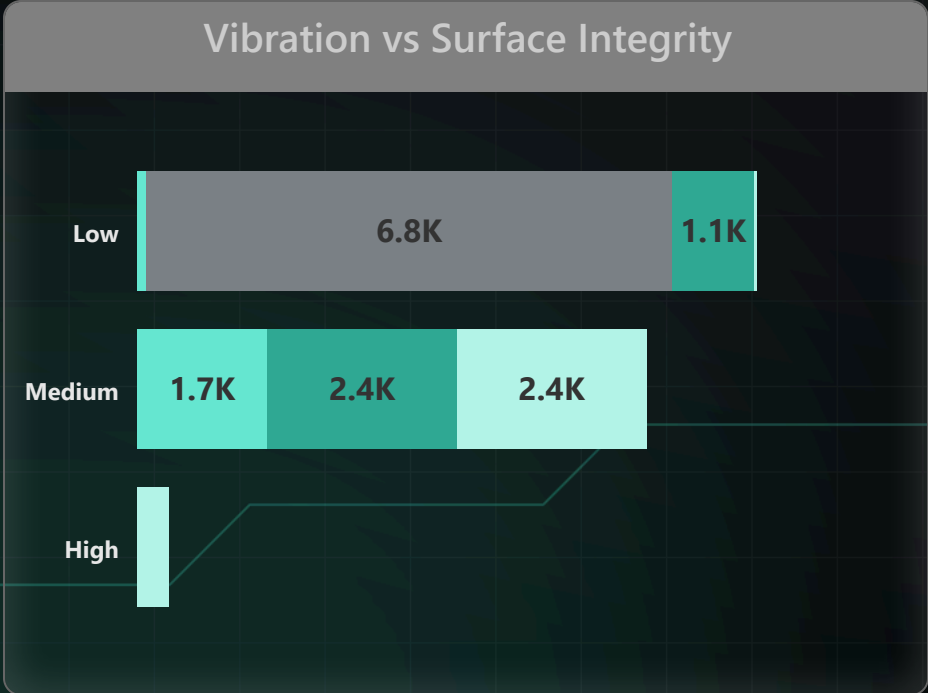
☐ Cold Chain Storage

☐ Distribution Center

☐ Retail Shelf

☐ Transport Vehicle

☐ Warehouse





Sample_ID	Food_Category	Packaging_Material	Storage_Zone	Package_Size	Food_Safety_Risk_Level	Compliance_Status	Digital_Twin_State	Sum of Temperature_C	Sum of Humidity_Percent	Sum of Oxygen
IPKG_09146	Fresh Produce	Aluminum Laminate	Cold Chain Storage	Bulk	Low Risk	Unsafe	Stable Twin State	6.46	39.87	
IPKG_06515	Meat Product	Biodegradable Film	Cold Chain Storage	Large	Low Risk	Unsafe	Stable Twin State	5.78	40.18	
IPKG_08324	Dairy	Glass Container	Cold Chain Storage	Small	Low Risk	Unsafe	Stable Twin State	3.07	40.91	
IPKG_05456	Seafood	Vacuum Pouch	Cold Chain Storage	Bulk	Medium Risk	Unsafe	Warning Twin State	5.19	41.11	
IPKG_06403	Beverage	Biodegradable Film	Cold Chain Storage	Large	Low Risk	Unsafe	Stable Twin State	4.18	41.34	
IPKG_02005	Frozen Food	Aluminum Laminate	Cold Chain Storage	Medium	Low Risk	Unsafe	Stable Twin State	4.34	42.42	
IPKG_12572	Beverage	Paperboard	Cold Chain Storage	Large	Low Risk	Unsafe	Stable Twin State	5.52	42.48	
IPKG_13114	Frozen Food	Vacuum Pouch	Cold Chain Storage	Medium	Low Risk	Unsafe	Stable Twin State	5.51	42.56	
IPKG_14102	Beverage	Composite Film	Cold Chain Storage	Small	Low Risk	Unsafe	Stable Twin State	3.64	42.84	
IPKG_00469	Seafood	Composite Film	Cold Chain Storage	Medium	Medium Risk	Unsafe	Warning Twin State	6.67	43.27	
IPKG_01703	Frozen Food	Glass Container	Cold Chain Storage	Large	Low Risk	Unsafe	Stable Twin State	6.22	43.31	
IPKG_14772	Beverage	Biodegradable Film	Cold Chain Storage	Small	Medium Risk	Unsafe	Warning Twin State	6.85	43.39	
IPKG_09098	Ready Meal	Composite Film	Cold Chain Storage	Small	Low Risk	Unsafe	Stable Twin State	5.08	43.43	
IPKG_03852	Seafood	PET Plastic	Cold Chain Storage	Bulk	Low Risk	Unsafe	Stable Twin State	3.95	43.52	
IPKG_12624	Meat Product	Paperboard	Cold Chain Storage	Medium	Low Risk	Unsafe	Stable Twin State	5.00	43.59	
IPKG_14127	Ready Meal	Glass Container	Cold Chain Storage	Large	Low Risk	Unsafe	Stable Twin State	4.12	43.61	
IPKG_10267	Canned Food	Paperboard	Cold Chain Storage	Large	High Risk	Unsafe	Alert Twin State	10.79	43.88	
IPKG_03322	Seafood	Aluminum Laminate	Cold Chain Storage	Bulk	Medium Risk	Unsafe	Warning Twin State	6.37	44.04	
IPKG_09007	Canned Food	PET Plastic	Cold Chain Storage	Bulk	Low Risk	Unsafe	Stable Twin State	5.04	44.18	
IPKG_10708	Ready Meal	Vacuum Pouch	Cold Chain Storage	Medium	Medium Risk	Unsafe	Warning Twin State	6.28	44.33	
IPKG_11510	Snack Food	Biodegradable Film	Cold Chain Storage	Large	Medium Risk	Unsafe	Warning Twin State	5.52	44.41	
Total								20,258.08	184,919.96	